REFLEXIVITY OF THE KERNEL OF A SURJECTION WITH REFLEXIVE MIDDLE TERM AND TORSION-FREE QUOTIENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Hom modules between finite modules over a Noetherian ring are finite. Algebra, Lemma \ref{algebra-lemma-ext-noetherian}, cited at proof lines 5263-5264. Makes the duals and double duals finite modules, so the diagram chase is a statement about finite modules.
- A finite module over a Noetherian domain is torsion free if and only if the map to its double dual is injective. Lemma \ref{lemma-reflexive-torsion-free}, cited at proof line 5272. Gives injectivity of the right vertical arrow of the diagram.
- Localisation at the fraction field is compatible with Hom out of a finitely presented module. Algebra, Lemma \ref{algebra-lemma-hom-from-finitely-presented}, cited at proof line 5288. Produces the exact sequence of vector spaces whose exactness forces Q tensored with the fraction field to vanish.

Source reference. Original lemma 0EB8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17913–17913. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08YR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11349–11349. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AV0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13668–13668. Navigation pointer to a statement the

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Source reference. Original lemma 0583: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6766–6766. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian domain. Let $0 \to M \to M' \to M''$ an exact sequence of finite R-modules. If M' is reflexive and M'' is torsion free, then M is reflexive.*

Proof. We will use without further mention that $\mathrm{Hom}_R(N, N')$ is a finite R-module for any finite R-modules N and N', see Algebra, Lemma 08YR. We take duals to get a sequence

$$\mathrm{Hom}_R(M, R) \leftarrow \mathrm{Hom}_R(M', R) \leftarrow \mathrm{Hom}_R(M'', R)$$

Dualizing again we obtain a commutative diagram

$$\begin{array}{ccccc} \mathrm{Hom}_R(\mathrm{Hom}_R(M,R),R) & \xrightarrow[j]{} & \mathrm{Hom}_R(\mathrm{Hom}_R(M',R),R) & \longrightarrow & \mathrm{Hom}_R(\mathrm{Hom}_R(M'',R),R) \\ \uparrow & & \uparrow & & \uparrow \\ M & \longrightarrow & M' & \longrightarrow & M'' \end{array}$$

We do not know the top row is exact. But, by assumption the middle vertical arrow is an isomorphism and the right vertical arrow is injective (Lemma 0AV0). We claim j is injective. Assuming the claim a diagram chase shows that the left vertical arrow is an isomorphism, i.e., M is reflexive.

Proof of the claim. Consider the exact sequence $\mathrm{Hom}_R(M', R) \to \mathrm{Hom}_R(M, R) \to Q \to 0$ defining Q. One applies Algebra, Lemma 0583 to obtain

$$\mathrm{Hom}_K(M' \otimes_R K, K) \to \mathrm{Hom}_K(M \otimes_R K, K) \to Q \otimes_R K \to 0$$

But $M \otimes_R K \to M' \otimes_R K$ is an injective map of vector spaces, hence split injective, so $Q \otimes_R K = 0$, that is, Q is torsion. Then one gets the exact sequence

$$0 \to \mathrm{Hom}_R(Q, R) \to \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R) \to \mathrm{Hom}_R(\mathrm{Hom}_R(M', R), R)$$

and $\mathrm{Hom}_R(Q, R) = 0$ because Q is torsion. □